

Bio-Friendly FRP Manufacturing: Jute-Based Composites and Bio-Resins as Eco-Friendly Materials for Sustainable Infrastructure in SMEs

Mostafa Shawkat Imran*, Abdullah Sayem**

Mostafa Shawkat Imran*, CEO, Nobo Shakti Prokushal, 73/H (GF) Green Road, Dhaka, Bangladesh,
E-mail: mostafa@noboshaktiprokushal.com

Abdullah Sayem**, Lecturer, Department of Fabric Engineering, Bangladesh University of Textiles,
Tejgaon IA, Dhaka, Bangladesh, E-mail: sayembutex@gmail.com

ABSTRACT

This study presents an end-to-end methodology for developing eco-friendly fiber-reinforced polymer (FRP) composites using locally sourced jute fibers and fully bio-based resin systems, targeting sustainable infrastructure applications in small and medium enterprises (SMEs). We synthesized a plant-oil-derived epoxy analogue and optimized its formulation through lab-scale mixing trials to balance viscosity (500–800 cP) and cure kinetics (gel time $\approx$ 60 min at 80 °C). Unidirectional jute mats were vacuum-infused and thermally cured to produce 300 × 300 × 5 mm panels.

Process consistency was ensured via an Internet of Things (IoT) monitoring network—comprising thermocouples, pressure transducers, and inline viscometers—which streamed real-time data to a cloud dashboard. Data-driven quality control reduced batch-to-batch variability by 35%. Artificial intelligence (AI) regression models, trained on 50 experimental runs, predicted flexural strength from in-process sensor readings with an R^2 of 0.91 in holdout tests.

Mechanical characterization (ASTM D790) demonstrated an average flexural strength of 45 MPa and impact resistance of 6 kJ/m², meeting or exceeding conventional glass-epoxy benchmarks for non-critical structural panels. Life cycle assessment revealed a 40% reduction in embodied carbon compared to petroleum-derived FRPs.

By integrating the resin formulation, IoT kit, and AI analytics into a modular SME-Ready Composite Lab, this work offers a scalable pathway for low-carbon FRP adoption in emerging markets.

Keywords: Jute fiber; Bio-resin; IoT monitoring